NLOS Identification and Ranging Trustworthiness for Indoor Positioning With LLM-Based UWB–IMU Fusion

Hongchao Yang, Yunjia Wang, Chee Kiat Seow, *Senior Member, IEEE*, Zengke Li, Meng Sun, Cedric De Cock, Jingxue Bi, Wout Joseph, *Senior Member, IEEE*, and David Plets, *Member, IEEE*

***Abstract*— In the rapidly evolving Internet of Things (IoT) landscape, accurate indoor positioning is increasingly vital. The proposed algorithm synergizes an ultrawideband (UWB) sensor with an inertial measurement unit (IMU) and artificial intelligence to obtain precise positioning in nonline-of-sight (NLOS) scenarios. In the proposed UWB module, a large language model (LLM) such as bidirectional encoder representations from transformers (BERT) algorithm is designed to utilize the channel impulse response (CIR) for effective NLOS identification and UWB ranging trustworthiness evaluation. Concurrently, the IMU module is also designed with BERT to recognize various pedestrian activity states, thereby optimizing positioning. BERT's self-attention mechanism and deep learning (DL) bidirectional training efficiently extract essential features from sequential data, capturing both local and global information. The integration of both UWB and IMU through a proposed tightly coupled algorithm significantly boosts positioning performance. Experimental campaigns demonstrate an average NLOS identification accuracy, line-of-sight (LOS), and $F2$ of 98.8%, 99.4%, and 0.9926, respectively. These performances surpass the state-of-the-art least-squares support vector machine (LS-SVM), convolutional neural network (CNN), and CNN with long short-term memory (CNN–LSTM) up to 17.66% in NLOS identification. In terms of pedestrian activity recognition using BERT, the BERT algorithm achieves a precision (recall) of 99.3% (99.4%), notably outperforming the CNN and CNN–LSTM by 17.9% (16.2%) and 11.9% (10.9%), respectively. Finally, the UWB–IMU algorithm significantly enhances positioning accuracy by 80.5%, outperforming Kalman, LSTM-EKF, and particle filter (PF) methods by 68.1%, 48.3%, and 45.0%, respectively. The proposed approach presents a robust solution for indoor positioning for IoT applications, particularly in challenging NLOS environments.**

***Index Terms*— Channel impulse response (CIR), indoor positioning system (IPS), inertial measurement unit (IMU), large language model (LLM), transformer, ultrawideband (UWB).**

Received 9 October 2024; revised 4 December 2024; accepted 7 January 2025. Date of publication 26 March 2025; date of current version 5 June 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2016YFB0502102 and in part by the National Natural Science Foundation of China under Grant 42304047. The Associate Editor coordinating the review process was Dr. Yulong Huang. *(Corresponding author: Hongchao Yang.)*

Hongchao Yang, Yunjia Wang, Zengke Li, and Meng Sun are with the Key Laboratory of Land Environment and Disaster Monitoring, China University of Ming and Technology, Xuzhou 221116, China (e-mail: hongchao_yang@cumt.edu.cn; wyjc411@163.com; zengkeli@yeah.net; msun@cumt.edu.cn).

Chee Kiat Seow is with the School of Computing Science, University of Glasgow, G12 8Rz Glasgow, U.K. (e-mail: CheeKiat.Seow@glasgow.ac.uk).

Cedric De Cock, Wout Joseph, and David Plets are with the imec-Waves Research Group, Department of Information Technology, Ghent University, 9000 Ghent, Belgium (e-mail: cedric.decock@ugent.be; wout.joseph@ugent.be; david.plets@ugent.be).

Jingxue Bi is with the School of Surveying and Geo-Informatics, Shandong Jianzhu University, Jinan 250101, China (e-mail: bijingxue19@sdjzu.edu.cn).

Digital Object Identifier 10.1109/TIM.2025.3554900

I. Introduction

LOCATION-BASED services (LBSs) are pivotal for diverse application domains, such as the Internet of Things (IoT) [1], virtual reality [2], and human–machine interactions [3]. While outdoor positioning is well-established using technology such as Global Navigation Satellite System (GNSS), indoor positioning systems (IPSs) are still evolving, holding significant market potential. Despite the availability of various indoor positioning technologies, many face challenges in complex, dynamic indoor environments. An ultrawideband (UWB) [4], [5] sensor, with its fine time resolution, stands out as a promising IPS. However, the UWB sensor also faces its challenge such as its time-of-flight (TOF)-based [6] ranging accuracy deteriorating in nonline-of-sight (NLOS) [4], [7] conditions, where its direct path between devices is obstructed. This results in positively biased errors, adversely affecting positioning accuracy and hindering seamless positioning. NLOS conditions are prevalent in intricate indoor settings, making the identification and mitigation of NLOS effects crucial for optimal positioning performance.

The UWB NLOS identification [8], [9] process typically involves channel feature extraction and classification. Classification methods include machine learning (ML) techniques like support vector machines (SVMs) [6], [10], decision tree [11], and generative adversarial networks (GANs) [12], as well as deep learning (DL) approaches such as convolutional neural networks (CNNs) [13], [14] and long short-term memory (LSTM) [5]. Features used are typically ranging distance [15], position [16], and channel impulse response (CIR) [5], [13]. Ranging-based methods that leverage UWB ranging distance may require extensive prior information and suffer from limited accuracy and threshold applicability. On the other hand, position-based methods using position features necessitate additional location data, increasing system complexity and reducing algorithm universality due to environmental constraints of the added technologies. The CIR sequence, generated during ranging, comprises the direct signal from the LOS path and multipath (MP) signals from reflection, refraction, and diffraction, containing critical environmental and obstacle information and is thus commonly used for NLOS identification. Besides using raw CIR values [5], [13], [14], NLOS identification methodology can also involve manually derived features obtained from CIR sequences' mathematical statistics [6], [8], [10]. It has been shown that combining derived CIR features with SVMs will yield an average accu-

1557-9662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

racy of 60%–70% [6], [10], but this performance fluctuates with environmental changes. Utilizing CIR's inherent channel information with DL's feature extraction capabilities can push NLOS identification accuracy above 80% [13], [14]. However, existing DL algorithms using CIR sequences struggle with issues such as overemphasis on local features, inefficient step-wise processing, and gradient descent vanishing or exploding problems that impede DL information flow. The purpose of NLOS identification is to distinguish between theoretically high-precision LOS and theoretically low-precision NLOS results. In NLOS scenarios where obstacles cause weak attenuation to the first-path (FP) signal, resulting in NLOS waveforms resembling LOS waveforms, or when the LOS FP signal fails to meet dynamic thresholds and exhibits NLOS characteristics, the CIR-based NLOS identification will fail. And the ranging accuracy in the above situation will be opposite to the theoretical channel environment. Therefore, it is not only necessary to improve the NLOS identification performance but also to further extract high-precision ranging results to ensure the accuracy of direct use for positioning data and leverage the advantages of UWB ranging and positioning.

In cases where identification outcomes are identified as NLOS with theoretically low-precision, several researchers have focused on improving ranging accuracy through the development of ranging mitigation models [17], [18]. This involves using ML (DL) algorithms to map the relationship between CIR features (CIR sequences) and ranging errors. However, the efficacy of CIR features is subject to environmental variability, requiring the re-selection of the most suitable CIR feature vectors for each environment. Integrating DL algorithms with CIR sequences is constrained by computational resources, posing challenges in capturing the dynamic relationships and trends within time-series data. Unifying the correction of NLOS errors can alleviate the impact of NLOS from the source. However, the adjusted ranging results may still not meet the requirements of the positioning algorithm, resulting in discontinuous positioning. In addition, it may cause some NLOS results that meet positioning requirements to be incorrectly corrected, resulting in increased errors and the inability to fully utilize the advantages of UWB ranging and positioning. Therefore, solely relying on UWB for positioning cannot fully mitigate the impact of NLOS conditions.

By introducing additional technologies and integrating UWB, the positioning performance under NLOS can be improved. Integrating wireless sensor Wi-Fi [19] or Bluetooth low-energy (BLE) [20], [21] fingerprinting and UWB can compensate for deficiencies in NLOS scenarios, enhancing positioning performance by 25%–45%. Although NLOS conditions enhance the uniqueness of fingerprints, constructing and maintaining a fingerprint database requires substantial resources. Emerging technologies like Wi-Fi fine timing measurement (WiFi-FTM) [22] and BLE angle of arrival (BLE-AOA) [23] can significantly improve the accuracy of hybrid systems. However, their ranging and angular measurement accuracy are also affected by NLOS phenomena. Introducing these additional wireless communication technologies can partially mitigate NLOS interference, but balancing precision, cost, and system stability remains challenging. Inertial measurement unit (IMU) such as pedestrian dead reckoning (PDR) [24], [25], [26], [27], [28], [29] offers an efficient, real-time solution unrestricted by indoor layouts through gait detection, step length estimation, and heading estimation, making it an ideal complement to UWB in NLOS scenarios. Researchers often use algorithms like Kalman filters (KFs) [29] and particle filters (PFs) [28] to fuse UWB and IMU positioning results [25], achieving an improvement of 40%–50% in positioning accuracy while ensuring real-time and continuous localization. The filtering algorithm is efficient and computationally lightweight but only utilizes current (t) and previous ($t-1$) information, resulting in limited information usage. In contrast, graph optimization models integrate UWB and IMU data into a positioning framework that accounts for data correlations over longer periods. This approach enhances the average positioning accuracy of IMU–UWB systems by over 50% [30], [31] and achieves an accuracy of 0.167 m when incorporating visual positioning [31]. However, existing fusion algorithms primarily emphasize direct weighting of the two positioning results during the combination and neglect the mutual information enhancement between the two technologies. For instance, improvements could include enhancing UWB channel identification and positioning accuracy or reducing IMU errors in step size and heading estimation. Additionally, current algorithms focus on suppressing rather than eliminating the effects of NLOS. Consequently, the low-precision results of UWB NLOS remain a significant factor in the final fusion positioning outcomes.

The large language model (LLM) such as bidirectional encoder representations from transformers (BERT) with a self-attention mechanism can automatically extract key features from sequential data. Its multihead self-attention mechanism concurrently focuses on multiple features, enhancing accuracy in classification tasks. BERT's deep bidirectional learning capability captures long-term dependencies within and across different sequences, improving efficiency through parallel processing. To address the aforementioned NLOS positioning challenges, this article proposed the use of the raw information of CIR sequences to combine with the BERT algorithm for NLOS identification. Based on the channel identification results, a trustworthiness evaluation for UWB ranging is proposed and established to maximize the retention of high-precision ranging data in both LOS and NLOS scenarios. To ensure positioning accuracy and continuity, an environment-independent IMU is further introduced to form a tightly coupled UWB–IMU positioning system. In the proposed system, UWB positioning results will correct the IMU's errors (stride, head, and position errors), while the IMU will act as a "pseudo"-UWB to assist in NLOS identification to replace low-precision NLOS ranging results. The contributions of this article are as follows.

1) We developed a trustworthiness evaluation system for UWB ranging results. To the best of our knowledge, this is the first introduction of such a model and concept. In this article, we enhance positioning accuracy by integrating NLOS identification with trustworthiness evaluation, extracting high-trustworthiness and corre-

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

sponding high-precision ranging results for positioning. This method maximizes the benefits of UWB signals, improving positioning precision while reducing the dependency of positioning performance on channel identification accuracy.

2) We utilize the self-attention-equipped BERT model for NLOS identification and pedestrian activity recognition. This article employs CIR sequences as input for NLOS identification, offering greater universality compared to the CIR feature vector method, which requires individual optimization for different scenarios. High-precision NLOS identification preserves accurate LOS ranging results and bolsters UWB ranging trustworthiness. BERT's bidirectional DL effectively utilizes multidimensional sensor data for precise pedestrian activity recognition, optimizing the step length model of the IMU algorithm and improving positioning accuracy.
3) We developed a tightly coupled positioning algorithm for UWB and IMU, aimed at achieving complementarity between the two submodules. This article utilizes IMU positioning results as "pseudo"-UWB to enhance UWB channel identification and address the issues of decreased accuracy and discontinuity in NLOS scenarios. Additionally, UWB positioning outcomes are employed to refine step length and heading estimates in the IMU system, correcting cumulative errors. Ultimately, the proposed algorithm is validated across various scenarios and multiple motion states.

The remainder of this article is organized as follows. Section II introduced related work of UWB NLOS identification and fusion positioning. Section III introduces the BERT and trustworthiness evaluation for UWB ranging. Section IV introduces the tightly coupled fusion of UWB and IMU. In Section V, we set up the experimental campaign in various scenarios to evaluate the algorithm's performance. Finally, Section VI gives some conclusions.

II. Related Work

A. NLOS Identification

NLOS phenomena disrupt UWB signal propagation along LOS paths, leading to positively biased errors in ranging results, which, in turn, degrade positioning accuracy and continuity. Therefore, NLOS identification is a focal point for UWB-based positioning systems. The simplest NLOS identification compares the probability density function (pdf) or variance of ranging results [8] against a fixed threshold to identify the channel environment [15]. However, these algorithms have limited accuracy and struggle to adapt to varying motion speeds and changing environments. The CIR sequence is fundamental to the UWB leading edge detection (LDE) [32] algorithm for identifying signal arrival times and has become a focal point in NLOS identification. CIR-based NLOS identification methods can be categorized into two approaches: ML with manually extracted features and DL utilizing the CIR sequence.

Manually extracted CIR features include mathematical statistical characteristics based on CIR sequence overall timing features (mean excess delay and kurtosis), energy features (first path power level and maximum CIR), and timing features related to signal arrival (rise time and first path error) [6], [10], [33]. Combining individual CIR features with SVMs [10] achieves channel identification accuracy of 60%–70% and a 70% LOS recall rate. However, performance can fluctuate over 10% due to changes in experimental scenarios or obstacles [33]. Compared to a single CIR feature, using a joint likelihood estimation algorithm [20] to combine multiple CIR features can increase identification accuracy by 5% and achieve over 85% accuracy under ideal thresholds. However, there is no universal answer regarding the number and composition of CIR features in the feature vector. In experiments by Khodjaev et al. [8], channel identification accuracy increased to 91.4% as the number of CIR features in the feature vector increased from 2 to 3. However, the introduction of a new CIR feature (maximum CIR) did not significantly alter accuracy. With three CIR features, Kolakowski and Modelski [34] improved channel identification accuracy by 22.8% by replacing specific CIR features. However, when the same feature vector from [8] was applied in another environment [18], channel identification accuracy showed significant fluctuations and even declined. Therefore, NLOS identification algorithms based on manually extracted CIR features require individual optimization for each environment. Additionally, they are more complex than algorithms using raw CIR sequence values due to additional feature extraction computations.

DL methods can automatically extract features from raw CIR sequence data to enable NLOS identification. The CNN combined with CIR sequences has achieved an average NLOS identification accuracy of 90% across various architectures, including ResNet and Encoder [14], [35]. However, the CNN primarily focuses on local features instead of global features. It may struggle to effectively capture channel characteristics in long CIR sequences. It can also process CIR sequence images, improving recognition accuracy by 33% when increasing image size from 64×64 to 256×256 [14]. However, data-to-image conversion requires more computational resources and increases latency. The CNN excels at extracting spatial information from input data, while the LSTM [5] is better suited for processing time-series information. When the number of hidden layers in the LSTM [5] increased from 1 to 2, accuracy improved by 18.5%. However, further increasing layers to 3 and 4 did not significantly change accuracy. The two-layer LSTM's identification accuracy was 1.1%, 12.4%, and 7.4% better than the two-layer Bi-LSTM, 1-D CNN, and gated recurrent unit (GRU), respectively. On the other hand, the two-layer Bi-LSTM outperforms the CNN and GRU by 11.3% and 6.3%, respectively. The LSTM cannot avoid the abundance of redundant information in long CIR sequences, slowing down training and leading to gradient anomalies that hinder training. Compared to NLOS identification algorithms based on CIR features, CIR sequence-based NLOS identification can identify NLOS data more accurately and intuitively, offering higher environmental robustness. However, existing CIR sequence-based NLOS identification algorithms struggle to balance the local and global information in sequence data. The step-wise processing of long CIR sequences or converting

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

CIR sequences into image data can lead to reduced operational efficiency and gradient anomalies due to the abundance of redundant CIR information, hindering effective information transfer.

The primary objective of NLOS identification is to isolate high-precision LOS results. However, existing algorithms fail to consider the presence of low-precision ranging results in LOS or high-precision results in NLOS, thereby underutilizing the advantages of UWB ranging and negatively impacting positioning performance. We proposed an NLOS identification algorithm based on BERT combined with the CIR sequence, which overcomes the inefficiencies and limitations of manual feature extraction. The deep bidirectional learning capability and the multihead self-attention mechanism of BERT address the constraints of traditional DL algorithms, which tend to focus on local features. Additionally, a trustworthiness evaluation algorithm is introduced to ensure the retention of high-precision ranging results, thereby enhancing positioning accuracy.

B. Fusion Positioning

Common algorithms that compensate for NLOS ranging errors by constructing error models require substantial computational resources and cannot guarantee that the corrected ranging will meet positioning requirements. Therefore, integrating additional technologies with UWB to enhance positioning performance can provide higher stability and feasibility. Additional technologies can be categorized into wireless and nonwireless signals.

Wireless signals, such as Wi-Fi [19] or BLE [21], enable fingerprint-based positioning, which is unaffected by NLOS conditions. In fact, NLOS signals can enhance the uniqueness of signal feature fingerprints, improving accuracy. However, due to the limited accuracy of fingerprint positioning (1–2 m), an ideal fusion system requires a sufficient number of UWB anchors to ensure a high proportion of LOS conditions. Wi-Fi FTM based on ranging [22] and BLE AOA [23] can achieve submeter-level positioning accuracy with a single base station fusion. These methods do not require the resource-intensive creation and maintenance of a fingerprint database, but they are affected by NLOS conditions. Therefore, the fusion of UWB with wireless communication technologies faces the challenge of balancing accuracy and the ability to work under NLOS interference.

Nonwireless signals, such as visual positioning systems, offer high precision and can incorporate additional map information to enhance positioning performance when combined with UWB information. Lin and Zhan [36] used the short-term positional changes of UWB to improve the attitude estimation of visual–inertial odometry, enhancing positioning accuracy by 20 cm while smoothing motion trajectories. Shi et al. [37] estimated the six degrees of freedom for vehicles and maps by minimizing UWB ranging errors and landmark reprojection errors, achieving a positioning accuracy of 0.025–0.036 m in public datasets. However, the accuracy of visual positioning significantly decreases in low-light conditions such as at night or underground, and the fusion algorithms require substantial computational resources. In contrast, IMU-based systems are less constrained by environmental factors and rely on simple, cost-effective sensors. IMU can improve the real-time positioning accuracy and continuity of UWB–IMU integrated systems under NLOS conditions. Positioning accuracy can be enhanced by 40%–50% using filtering algorithms [27], [28], [29]. While IMU offers high short-term accuracy, its step-length estimation struggles to adapt to varying pedestrian activities, leading to the rapid accumulation of positioning errors and negatively affecting fusion accuracy. Therefore, accurately identifying pedestrian activities [38], [39] to adjust the step-length model is crucial for improving the accuracy of IMU positioning. Susi et al. [40] used low-pass filtering for sensor data preprocessing and recognized four motion states using decision tree classification, achieving an accuracy of 95%. Although filtering can reduce noise, it requires sufficiently long sequences to preserve the fluctuations of the original data. Bouchut et al. [41] manually extracted time and energy features from accelerometer data and used SVMs to identify six activity states, achieving an accuracy of 97.5%. This algorithm can be embedded in hardware for real-time classification, but it can only differentiate between one activity and another. Bashar et al. [42] used neighborhood component analysis and neural networks (NNs) to select key features, obtaining an accuracy of 95.8% on public datasets. Zebhi [43] transformed sensor data into 2-D images using a 2-D fast Fourier transform and combined this with the CNN to achieve an accuracy of 93.5%–99.0%. However, this image transformation increases latency and computational demands. Current algorithms struggle to extract sufficient activity recognition feature information from raw sensor data without considering the fluctuations of multidimensional sensors.

IMU systems are not affected by indoor environment propagation anomalies. They provide short-term, high-precision, and continuous positioning results. Thus, the fusion of IMU and UWB could fundamentally alleviate NLOS interference. Xia et al. [25] developed an NLOS accuracy factor model based on the distance difference between UWB and IMU, using IMU positioning only when the accuracy factor exceeded a predefined threshold. Results showed an average increase in positioning accuracy of 53.4% and a reduction in standard deviation by 76.3%. However, the threshold definition was unclear and required adjustment according to scenario changes. Vandermeeren and Steendam [28] used filtering techniques for IMU and UWB fusion positioning of supermarket shopping carts. Standalone UWB accuracy was 0.6 m, but achieved 0.3 and 0.4 m accuracy with KF and PF-based UWB–IMU algorithms, respectively. However, the orientation of IMU sensors on shopping carts is more stable than on pedestrians, and the experimental environment lacked additional human interference, resulting in lower NLOS occurrence. Guo et al. [26] employed a variable noise variance KF to fuse UWB and IMU positioning, effectively reducing reliance on noise statistical distribution and improving accuracy by 67.7% over KF. However, pedestrian motion trajectories and states were relatively uniform. Ali et al. [29] used least-squares (LS) and adaptive KF for real-time compensation of UWB and IMU positioning errors, comparing their performance in tightly

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

and loosely coupled scenarios. Tightly coupled UWB–IMU improved positioning accuracy by 53.3% to 0.11 m compared to loosely coupled. The position estimation of the filtering algorithm relies solely on the current and previous state information, which lacks the long-term memory of the state. In contrast, the graph optimization algorithm [30], [31], [44] incorporates historical and current measurement values to estimate the current position and improve positioning accuracy under constraint conditions. Considering the characteristics of UWB ranging and planar motion, this article introduces UWB ranging constraints and planar motion constraints, focusing on relevant measurements and residual construction in the model, to achieve a 24.52% improvement in positioning accuracy [31]. Building on this, the study in [30] uses the communication ranging results between UWB anchors as a joint graph factor to improve the fusion positioning performance of multiple anchors. Increasing the number of base stations from 5 to 7 improves the graph optimization algorithm's positioning accuracy by 49.18%. During the base station expansion, the graph optimization algorithm's accuracy, when compared to the PF algorithm, has outperformed from 12.11% to 25.41%. Existing algorithms typically combine individual positioning results from IMU and UWB to improve fusion performance and continuity. However, when the number of UWB devices is insufficient to meet the required positioning accuracy, the fusion algorithm will depend entirely on IMU. This will significantly reduce overall system accuracy.

In contrast, the proposed fusion algorithm uses UWB positioning results to optimize IMU step length and heading accuracies to address the IMU's cumulative errors. Additionally, the proposed algorithm treats IMU positioning results as UWB pseudo-anchors, aiding both NLOS identification and UWB positioning under NLOS conditions. The proposed UWB–IMU algorithm, leveraging BERT, resolves IMU fluctuations through activity recognition enabled by BERT. Furthermore, the tightly coupled integration of BERT for both NLOS identification and IMU activity recognition enhances positioning accuracy beyond the performance of each method individually.

III. Background

UWB technology can achieve centimeter-level ranging accuracy through its high temporal resolution. The ability to work under MP interference enables the distinction between true distance signals from LOS paths and reflected signals from other paths. The built-in LDE algorithm [32] in UWB systems can differentiate between UWB signals and environmental noise, using CIR data [10] to identify the signal arrival time. The CIR is formulated as follows:

$$r(t) = \sum_{j=1}^{J} \alpha_j p\left(t - \tau_j\right) + \eta(t) \tag{1}$$

where α_j and τ_j are the amplitude and time delay of the jth path, respectively, $p(\cdot)$ denotes the received UWB pulse, and $\eta(t)$ is the additive white Gaussian noise (AWGN) with zero mean. The LDE algorithm dynamically calculates a threshold based on real-time CIR samples. The first CIR sample exceeding this threshold is identified as the FP signal, and its corresponding time is recorded as the signal arrival time. Due to MP effects, the sequence also contains the strongest path (SP), characterized by the maximum CIR value. Fig. 1 depicts a typical LOS (NLOS) CIR sequence. In this study, the CIR segment preceding the FP (blue in Fig. 1) is defined as the environment noise stage (ENS). The CIR between the FP and SP signals (orange in Fig. 1) is termed the steep rise stage (SRS), and the subsequent CIR (green in Fig. 1) is categorized as the slow descent stage (SDS).

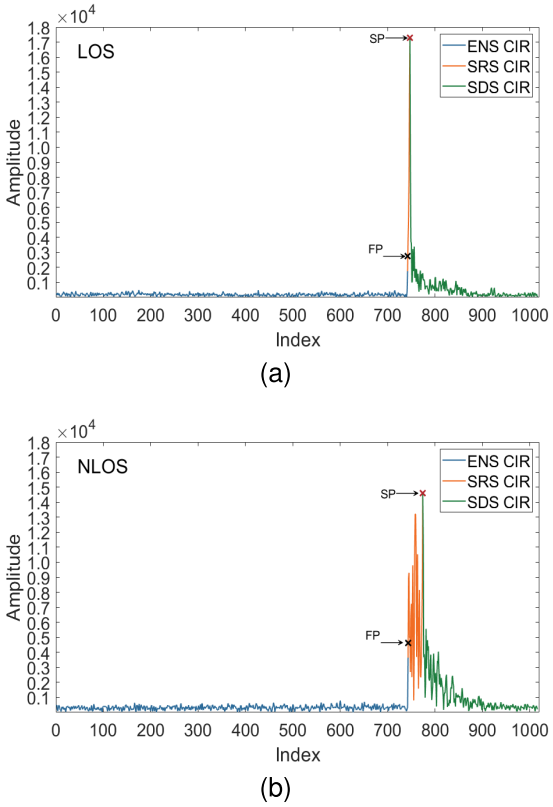


Fig. 1. Typical UWB CIR sequence in different channel environments. (a) LOS. (b) NLOS.

In NLOS environments, obstacles obstruct and attenuate the energy of the true FP signal, making it lower than the threshold of the LDE algorithm. Consequently, the LDE algorithm erroneously classifies high-energy UWB CIR samples as ENS and identifies subsequent MP signals as FP. As shown in Fig. 1, the CIR energy in NLOS ENS after the 600th sample is significantly higher than that in the corresponding LOS ENS. The characteristic of SRS is due to the energy surge caused by UWB arrival. Under LOS conditions, the CIR of SRS quickly reaches its maximum value, followed by SDS, which is the gradually attenuating MP signal. However, under NLOS conditions, pronounced MP effects delay the peak of the SP, resulting in higher fluctuations in the CIR of subsequent SDS compared to LOS situations. Thus, the classification algorithm only needs to combine the aforementioned segments in the CIR sequence to achieve high-precision NLOS identification. This approach avoids the complexity of computing CIR features and the optimization of CIR feature vectors for different environments.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

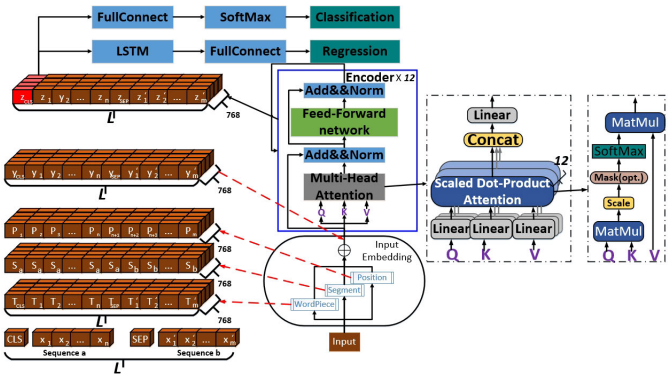


Fig. 2. Flowchart of BERT for classification.

A. Bidirectional Encoder Representations From Transformers

BERT [45] incorporates a self-attention mechanism [46] that employs multihead self-attention computation to focus on features in long sequences, as illustrated in Fig. 2. It achieves deep bidirectional learning through two pretraining tasks: masked language modeling (MLM) and next sentence prediction (NSP), enabling the simultaneous capture of both local features and global relationships within sequence information, as well as relationships between different sequences. Consequently, compared to DL algorithms, which are limited to local features, BERT extracts more comprehensive and superior information. Additionally, its parallel processing capability enables efficient processing of information across all positions in the sequence, thereby enhancing overall efficiency.

For the input containing multiple independent sequences, BERT utilizes special symbols [CLS] to mark the beginning and [SEP] to segment the sequences. For an input vector X of length L, BERT employs input embedding (WordPiece+Segment+Position) to convert it into the input matrix Y for the encoder stack. WordPiece encodes the CIR sequence and multidimensional sensor information based on the corpus, while Segment labels identify the sequence to which each piece of information belongs. As BERT does not need recursion and convolution, it uses "position embedding" to signify data positions and relative distance of data within the vector. WordPiece, Segment, and Position first convert X into three vectors of length L and expand them into three matrices ($L \times d_{\text{emd}}(=768)$) through deep bidirectional learning of BERT, named T, S, and P, respectively. For NLOS identification using a CIR sequence (1 sequence), BERT can fully explore the correlation features between local and global CIR samples in CIR sequences. For the trustworthiness evaluation based on the CIR sequence and the ranging result (2 sequences) as inputs, BERT can further obtain the correlation information between the ranging results and each CIR sample. For the pedestrian activity recognition based on triaxial acceleration and overall acceleration (four sequences) as input, BERT can not only obtain the local global characteristics of a single sensor axis but also obtain the simultaneous fluctuation information of different sensors at the same time. The above information is stored in T, S, and P through embedding and overlaid to form $Y \in \mathbb{R}^{L\times d_{\text{emd}}}$ as the input of the encoder stack. BERT uses the self-attention mechanism in the encoder stack to extract and enhance the weight of important information (channel information, trustworthiness information, and pedestrian activity information) in Y according to task requirements to achieve different goals.

As shown in Fig. 2, the encoder stack comprises $C = 12$ identical encoders, each containing two sublayers: a multihead self-attention mechanism (Multihead) and position-wise fully connected feed-forward network (FFN). The multihead attention consists of $D = 12$ Scaled Dot-Product Attention units, where the inputs are queries (Q), keys (K), and values (V), operating under the following formula:

$$Q_d = Y * W_d^Q \quad Q_d \in \mathbb{R}^{L\times d_k} \tag{2a}$$

$$K_d = Y * W_d^K \quad K_d \in \mathbb{R}^{L\times d_k} \tag{2b}$$

$$V_d = Y * W_d^V \quad V_d \in \mathbb{R}^{L\times d_v} \tag{2c}$$

where $d = 1, \ldots, D$, and the projections are parameter matrices $W_d^Q \in \mathbb{R}^{d_{\text{emd}}\times d_k}$, $W_d^K \in \mathbb{R}^{d_{\text{emd}}\times d_k}$, and $W_d^V \in \mathbb{R}^{d_{\text{emd}}\times d_v}$, $d_k = d_v = d_{\text{emd}}/C = 64$. Scaled dot product attention is

$$\text{Attention}_d(Q_d, K_d, V_d) = \text{softmax}\left(\frac{Q_d \times K_d^T}{\sqrt{d_k}}\right)V_d. \tag{3}$$

After the MatMul operation, the combination of Q_d and K_d forms an attention map matrix of size $L \times L$. The Mask(opt.) operation eliminates the weight information from the upper right corner of the attention map matrix. This ensures that each piece of data is only aware of its own attention weight and that of the preceding data. The scaling operation [MatMul, Mask(opt.)] involves dividing the data by $\sqrt{d_k}$ to scale the matrix, preventing the subsequent SoftMax function from entering the region of minimal gradients. The purpose of SoftMax is to normalize the weights so that their sum equals 1. After obtaining the attention weight matrix with Q_d and K_d, it is combined with V_d to produce the final matrix $\text{Attention}_d \in \mathbb{R}^{L\times d_v}$. The result of Multihead, which consists of 12 Scaled Dot-Product Attention, is as follows:

$$\text{Multihead}_c(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_{12})W_c^O \tag{4a}$$

where

$$\text{head}_d = \text{Attention}_d(Q_d, K_d, V_d) \tag{4b}$$

where $W_c^O \in \mathbb{R}^{(D\times d_v)\times d_{\text{emd}}}$ is the parameter matrix of the cth encoder, where $c = 1, \ldots, C$ and $\text{Multihead}_c \in \mathbb{R}^{L\times d_{\text{emd}}}$. During the learning process, by adjusting the weight matrix, the model can focus more on information with higher weights that affect the target task. The subsequent layer after Multihead Attention is a layer normalization (Add&&Norm) with residual connection [47]. In the cth encoder, the output of first Add&&Norm is $\text{LayerNorm}_c^1 = \text{Norm}(Y + \text{Multihead}_c(Q, K, V))$. This output is fetched to the next sublayer of the encoder which is a fully connected feed-forward NN (FNN) that consists of two linear transformations connected by ReLU activation, described by the following equation:

$$\text{FFN}_c\left(\text{LayerNorm}_c^1\right) = \max\left(0, \text{LayerNorm}_c^1 W_1 + b_1\right)W_2 + b_2 \tag{5}$$

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

where W_1 and W_2 are size of $d_{\text{emd}} \times d_{\text{ff}}(=3072)$ and $d_{\text{ff}} \times d_{\text{emd}}$, respectively. The next layer after FNN is the second Add&&Norm with the output $\text{LayerNorm}_c^2 = \text{Norm}(\text{FFN}_c + \text{LayerNorm}_c^1)$. After the input vector Y passes through $C = 12$ encoders, we obtain the transformed vector $Z \in \mathbb{R}^{L \times d_{\text{emd}}}$, which contains a special token Z_{CLS}. In fact, Z is the transformed data that provides the embodiment of the distinct feature characteristics of channel information and ranging trustworthiness for the CIR sequence and ranging, and IMU activity for the IMU sequence input. Furthermore, Z_{CLS} token is pivotal, serving not just as a marker for the entire input data but also as a representative of the entire input's information. Consequently, the vector of the Z_{CLS} token $\in \mathbb{R}^{1 \times d_{\text{emd}}}$ encapsulates feature information (either NLOS/LOS characteristics, ranging trustworthiness, or IMU activity) of the entire Z sequence. For classification tasks, the feature information of Z_{CLS} is passed through a fully connected NN and Softmax layers to lead to NLOS identification and activity recognition. In the case of regression tasks, the features of Z_{CLS} are processed through an LSTM layer, a fully connected NN, and regression layers to generate the target output for ranging trustworthiness evaluation.

B. IMU Positioning

Utilizing sensor-acquired acceleration (Acc) data, gyroscope (Gyr) data, and orientation angle (Ori) data, IMU estimates position through gait detection, step length estimation, and heading estimation. The Acc data comes from the original three-axis accelerometer [X(Acc_x), Y(Acc_y), and Z(Acc_z)] with the following formula:

$$\text{Acc} = \sqrt{\text{Acc}_x^2 + \text{Acc}_y^2 + \text{Acc}_z^2}. \tag{6}$$

Acc data is preprocessed using a bandpass filter, and gait detection is accomplished based on peak detection. The purpose of gait detection is to segment the sequence Acc and calculate the movement distance corresponding to each segment, that is, step-length estimation, as defined by the following formula:

$$\text{Stride} = a + b * \text{StepDur} + c * \text{StepAv} \tag{7}$$

where Stride is the length of a step, StepDur is the duration of a step, StepAv is the variance of sequence Acc within the step duration, and a, b, c are the fitting parameters. Stride estimation is an offline training that can be adapted to different user profiles [26]. For heading estimation, we integrate the rotational velocity of the three-axis Gyr to estimate the device's orientation (Head). For Stride and direction Head estimation, we can update the position relative to the previous moment and calculate the coordinate changes using the rotation matrix [28]. It is noteworthy to state that our proposed BERT for activity recognition can be adapted to different users as it can recognize different use profiles which is one of our proposed novelties using BERT. As such, to the best of our knowledge, we are the first to employ BERT for pedestrian activity recognition, followed by the calculation of StepLength parameters based on different pedestrian activity states. The relevant parameters of the gait model based on different motion states are collected during the offline stage. Therefore, the characteristics of different pedestrians (e.g., height, gender, etc.) collected during offline training will form the different parameters into our proposed BERT during BERT offline training to facilitate model recognition and also user profile recognition.

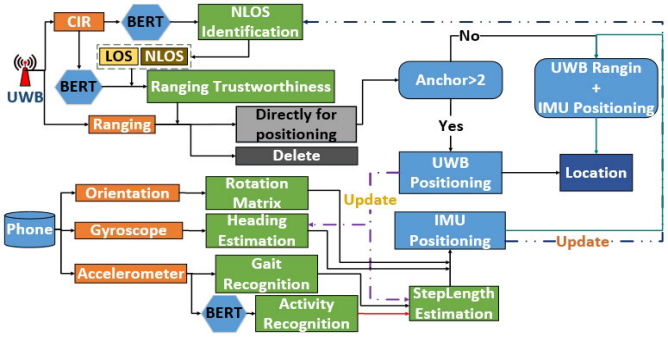


Fig. 3. Structure of UWB is tightly coupled with IMU.

IV. Proposed Method

Fig. 3 illustrates the proposed tightly coupled framework of UWB and IMU. In the UWB module, we initially use the CIR sequence in conjunction with BERT for NLOS identification. From the NLOS channel identification results, we determine the trustworthiness of ranging results using BERT. Results deemed low in trustworthiness are discarded, while those with high trustworthiness are used directly for positioning. In the IMU module, we use Acc data for gait recognition and combine it with the BERT that was used for pedestrian activity recognition to select the appropriate step-length model. The integration of Gyr data is used to estimate the pedestrian's heading and coupled with step length to infer IMU's relative displacement. Finally, the displacement from different pedestrian activities is standardized into the same coordinate system using the rotation matrix based on Ori.

During the fusion positioning stage, the strategy proposed here goes beyond merely filtering and fusing different positioning results. It constructs a complementary fusion approach between UWB and IMU submodules. When there are more than three highly trustworthiness ranging results, UWB employs an LS method for calculating positioning results. Simultaneously, these results refine the step size and heading estimates of the IMU, thus enhancing IMU positioning accuracy. The real-time, continuous IMU positioning results act as UWB pseudo-anchors to assist in UWB's NLOS identification. Furthermore, when the number of trustworthiness ranging results is insufficient for precise positioning requirements, IMU position serves as supplementary data for fusion positioning. Subsequently, this article will introduce methodologies for UWB ranging trustworthiness evaluation, IMU pedestrian activity recognition, and fusion positioning algorithms.

A. Trustworthiness Evaluation for UWB Ranging

Due to its small pulsewidth or high bandwidth, UWB signals have high resolution that can allow some NLOS ranging

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

signals to be used to meet positioning requirements. Complete elimination of useful NLOS signals can undermine the advantages of using UWB signals while attempting to correct them can decrease the ranging accuracy of the aforementioned data. Due to the presence of misclassification, data labeled as LOS after the BERT NLOS classification stage might include misclassified NLOS data and inherently low-precision LOS ranging results. Using all LOS-labeled data directly for positioning can reduce positioning performance. As such, we propose a novel trustworthiness assessment approach for UWB ranging results to assess ranging quality and further extract high-precision ranging results based on channel recognition. The trustworthiness of UWB ranging is defined as the probability that the ranging result to be less than 1 m and can be directly used for 2-D positioning, with a reliability score ranging from 0 to 1 for each result. A higher trustworthiness score indicates greater ranging accuracy.

Due to challenges in collecting real-world data with varying error levels, this study utilizes simulation experiments to evaluate the impact of ranging errors on positioning accuracy. Within a 100 × 100 m simulated area, three UWB anchor nodes and one target tag are randomly distributed within the environment for a 2-D positioning [48], [49]. In actual scenarios, only three anchor nodes will be used for positioning with LS. The actual distances from the target tag to each anchor node are then calculated. In the simulated experiment, a measurement error ranging from 0.001 to 5 m is incrementally added to one of the ranging results of an anchor node at intervals of 0.001 m, while the other two anchor nodes' results retain their true ranging distance to the target tag. This amounts to 5000 sets of data. The process is then repeated to extend to the remaining two anchor nodes, yielding a total of 15 000 unique sets of data. Each set comprises two actual ranging results and one ranging error. We repeated the whole process with different random UWB anchor nodes and target nodes distribution within the environment 500 times. At the end of the simulated experiments, we will gather 7.5 million datasets. The trustworthiness of the ranging error will then be defined as the percentage of positioning errors less than 1 m relative to the ranging error. The simulation data with the associated curve fitting using the obtained regression equation are shown in Fig. 4, where the horizontal axis denotes the ranging error and the vertical axis indicates the trustworthiness of these errors.

We propose the mapping of the real measurement ranging error to a trustworthiness score that is obtained in the above-mentioned simulation experiments, effectively illustrating the influence of ranging error on positioning accuracy. In the measurement experiment, high-precision visual positioning is used to ascertain the ground truth location of the anchor nodes and target tag under measurement, hence the true distance between them. The ranging error is calculated using the true distance and the measured ranging results. Hence, the trustworthiness is obtained by mapping to the regression equation obtained in Fig. 3. To assess the trustworthiness of ranging results without prior knowledge of the true ranging error, this article proposes the following ranging trustworthiness evaluation strategy.

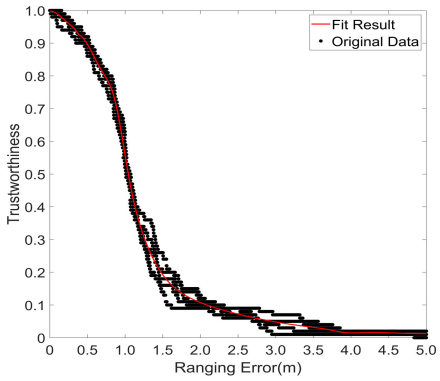


Fig. 4. Trustworthiness evaluation function of UWB ranging errors.

TABLE I
CLASSIFICATION FOR TRUSTWORTHINESS EVALUATION

	LOS		NLOS
Type1	$DRF < Thre_1$ and $rmse \leq Thre_2$	Type1	$DRF < Thre_1$
Type2	$k \geq Thre_3$	Type2	$k \geq Thre_4$
Type3	Other	Type3	Other

First, the traditional CIR features [10] [root mean square error (RMSE), kurtosis (k), and the difference between the receive signal level and the first path power level (DRF)] were utilized to classify the NLOS identification results from the first stage of BERT. This novel approach categorizes LOS and NLOS data into three distinct groups, as detailed in Table I. Subsequently, for each of the six categories, we will activate a 2-D feature vector, comprising the CIR sequence as sequence A and the ranging result as sequence B, to be input into a new BERT to construct a regression model that assesses the trustworthiness of ranging results. At the next stage, we will prioritize the ranging results with high trustworthiness for positioning purposes, while discarding those with low trustworthiness. This method, in contrast to traditional ranging error correction algorithms, obviates the need for a complex and environmentally sensitive mapping between CIR features and ranging errors. It offers reduced system complexity and minimizes the impact of channel identification accuracy on positioning performance. To further enhance the robustness of the UWB performance robustness and resiliency in different environment dynamics, we integrate and tightly coupled IMU technology, which remains unaffected by environmental factors and NLOS interference.

B. Activity Recognition

The real-time positioning of the IMU can effectively compensate for the performance degradation of UWB under NLOS conditions. However, varying pedestrian motion states can alter the step size pattern, leading to a rapid accumulation of IMU positioning errors and decreased accuracy in both individual and combined positioning systems. To address this, we have developed a recognition model for four common pedestrian activities when carrying a mobile phone:

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PSEUDO-CODE OF THE TIGHTLY COUPLED UWB AND IMU

Input: $\{time_n\}_{n=1}^{N=M+J}$; $\{Pos_m^{uwb}, Channel_i^m, d_i^m, Cred_i^m\}_{i=1,m=1}^{4,M}$; $\{Pos_j^{IMU}, Head_j, Stride_j\}_{j=1}^{J}$;
Output: Update $\{Pos_m^{uwb}\}_{m=1}^{M}$ and $\{Pos_j^{IMU}\}_{j=1}^{J}$.

$m = 1, j = 1$;
for $n = 3 : N$
  if $time_n$ is UWB position
    $m = m + 1$;
    update UWB Channel
    $\{Channel_i^m\}_{i=1}^{4} = \boldsymbol{Update_{LOS}}(Pos_j^{IMU}, Pos_m^{uwb}, \{Channel_i^m, d_i^m, \}_{i=1}^{4})$;
    update UWB Position
      $Pos_m^{uwb} = \boldsymbol{Fusion}(Pos_j^{IMU}, \{Pos_m^{uwb}\}_{m=1}^{m}, \{d_i^m, Cred_i^m\}_{i=1}^{4})$;
    update IMU position
    if $j \geq 2$ && $Num(Cred_i^m > Cred_{thre}) \geq 3$
      $Head_j$ is the heading from Pos_{j-1}^{IMU} to Pos_m^{uwb};
      $Stride_j$ based on the speed from Pos_{j-1}^{IMU} to Pos_m^{uwb};
      Update $\{Pos_j^{IMU}\}_j^J$;
    end
  elseif $time_n$ is IMU
    $j = j + 1$;
    $Stride_j$ based on the speed from Pos_{j-1}^{IMU} to high trustworthiness Pos^{uwb};
    if $time_{n-1}$ is UWB
      $Head_j$ is the heading from Pos_{j-1}^{IMU} to Pos_m^{uwb};
    end
    Update $\{Pos_j^{IMU}\}_j^J$
  end
end

1) holding the device at head level during a call (Call); 2) holding the device horizontally at chest level (Hold); 3) placing the device in the pocket of the right thigh (Pocket); and 4) swinging the device back and forth with the right hand (Swing). To the best of our knowledge, we are the first to employ BERT for pedestrian activity recognition. Leveraging BERT's deep bidirectional learning capability, we utilize a combination of overall acceleration (Acc) and tri-axial acceleration sequences (Acc_x, Acc_y, Acc_z) as inputs. BERT can fully extract the fluctuation patterns of three-axis acceleration itself and their correlations under different motion states while deepening the correlation with overall acceleration. To facilitate real-time motion state detection, we segment each set of sequence data. The data length is set to 30, with a sliding window shift of five data points for each step. Therefore, each activity recognition input comprises a full sensor sequence data with a length 4*30 comprising of an Acc, Acc_x, Acc_y, and Acc_z sequence. It is noteworthy to mention that our proposed BERT for activity recognition can be adapted to different users as it can recognize different user profiles which is one of our proposed novelties using BERT. The relevant parameters of the gait model specified in (7) based on different motion states for each user are collected during the offline stage. Therefore, the characteristics of different pedestrians (e.g., height, gender, etc.) collected during offline training will form the different parameters into our proposed BERT during BERT offline training to facilitate accurate identification of not only each pedestrian state recognition but also each user profile recognition.

C. IMU–UWB Fusion Position

1) Data Preprocessing: Table II presents pseudocode for the tightly coupled UWB and IMU systems, while Table III details the corresponding auxiliary functions. For M sets of UWB data where $m = 1, \ldots, M$ and $i = 1, \ldots, 4$ for four anchors, each set includes ranging results $\{d_i^m\}_{i=1,m=1}^{4,M}$ and CIR sequence from four anchors. Following the steps in Fig. 3, we utilize the CIR sequence and BERT to identify the channel state ($\{\text{Channel}_i^m\}_{i=1,m=1}^{4,M}$) and calculate the trustworthiness ($\{\text{Cred}_i^m\}_{i=1,m=1}^{4,M}$) of the ranging results. When there are three ranging results d_i^m with trustworthiness higher than $\text{Cred}_{\text{thre}}$, these are combined with the corresponding UWB anchor coordinates and input into LS to obtain UWB positioning results ($\{\text{Pos}_m^{\text{uwb}}\}_{m=1}^{M}$). If there are insufficient high-trustworthiness ranging results, $\{\text{Pos}_m^{\text{uwb}}\}_{m=1}^{M}$ are calculated using four ranging results. For J sets of IMU data corresponding to $J-1$ steps, each set includes position, step length, and heading ($\{\text{Pos}_j^{\text{IMU}}, \text{Stride}_j, \text{Head}_j\}_{j=1}^{J}$). The step-length model is adaptively switched based on the results of pedestrian activity recognition. $\text{Head}_1 = \text{Stride}_1 = 0$ and $\text{Pos}_1^{\text{IMU}}$ is obtained from the UWB position. We arrange M sets of UWB data and J sets of IMU data based on the time ($\{time\}_{n=1}^{N}$, $N = M+J$) to form a sequence of positioning information with a length of N. The UWB–IMU fusion algorithm begins with the continuous occurrence of UWB and IMU positioning results ($n = 3$ after $m = 1$ and $j = 1$), which is easily achieved based on the continuous output of IMU.

2) IMU Update UWB Submodule in UWB–IMU Fusion: As shown in Table II, when this moment (time_n) is UWB positioning, we treat the previous IMU position result ($\text{Pos}_j^{\text{IMU}}$) as a pseudo-UWB anchor to update the NLOS identification and position result of UWB. The ranging result (Dis) of pseudo-UWB anchor is the distance from $\text{Pos}_j^{\text{IMU}}$ to current $\text{Pos}_m^{\text{uwb}}$. Due to the ranging error, a circle centered at $\text{Pos}_j^{\text{IMU}}$ with a radius of Dis intersects with another circle centered at

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PSEUDO-CODE OF THE FUNCTION

Funtion $\{Channel_i^m\}_{i=1}^4 = \boldsymbol{Update_{LOS}}(Pos_j^{IMU}, Pos_m^{uwb}, \{Channel_i^m, d_i^m\}_{i=1}^4)$
Calculate the distance (Dis) between Pos_m^{uwb} and Pos_j^{IMU};
Calculate individually the number of intersection points ($\{Num_i\}_{i=1}^4$) between a circle centered at Pos_j^{IMU} with a radius of Dis_{tre} and the four circles each centered at the coordinate of UWB anchors with a radius of $\{d_i^m\}_{i=1}^4$;
Switch $sum(\{Num_i\}_{i=1}^4)$
Case 0 $\{Channel_i^m\}_{i=1}^4 = LOS$;
Case 2 $Channel_i^m = LOS$ for $Num_i = 0$; $Channel_i^m = NLOS$ for $Num_i = 2$;
Else $\{Channel_i^m\}_{i=1}^4 = \{Channel_i^m\}_{i=1}^4$;
end
Funtion $Pos_m^{uwb} = \boldsymbol{Fusion}(Pos_j^{IMU}, \{Pos_m^{uwb}\}_{m=1}^m, \{d_i^m, Cred_i^m\}_{i=1}^4)$
Find the highest trustworthiness Pos^{uwb} in $\{Pos_m^{uwb}\}_{m=1}^m$;
Calculate the speed between Pos^{uwb} and Pos_j^{IMU} and corresponding distance (d_{IMU}) from time of Pos_j^{IMU} to Pos_m^{uwb};
Switch $Num(Cred_i^m > Cred_{thre}))$
Case 3&4
Pos_m^{uwb} is the LS with ranging (d_i^m) and the coordinate of UWB anchors;
Case 2
Pos_m^{uwb} is the LS with ranging (d_i^m, d_{IMU}) and the coordinate of two high-trustworthiness UWB anchors and Pos_j^{IMU};
Case 1
Pos_m^{uwb} is the mean coordinate of the intersection for two circles with Pos_j^{IMU} and Pos_i^{Ac} as centers, and d_{IMU} and d_i^m as corresponding radii;
Case 0
Pos_m^{uwb} is equivalent to move one step from Pos_j^{IMU}, with $Stride = d_{IMU}$ and $Head$ is extension of Pos_j^{IMU} and Pos^{uwb};
end

the coordinate of the UWB anchor with a radius of d_i^m at two points. In the LOS condition, a slight reduction in the radius Dis is sufficient to eliminate the intersection points. This can serve as a checkpoint to determine any larger ranging error that will occur in NLOS situations. As such, we use Dis_{thre} as the radius for the circle $\text{Pos}_j^{\text{IMU}}$ to calculate the number of intersection points with the UWB anchor circle and establish corresponding channel update rules as shown in Table III under $\text{Update}_{\text{LOS}}$. It is important to note that an update in Channel_i^m will result in a corresponding update in Cred_i^m.

Next, we update the UWB position. When $\text{Num}(\text{Cred}_i^m > \text{Cred}_{\text{thre}}) \geq 3$, we combine their d_i^m and LS to calculate $\text{Pos}_m^{\text{uwb}}$. When there are only two or fewer high-trustworthiness UWB ranging results, we will treat IMU as a pseudo-UWB anchor, and the ranging result (d_{IMU}) is based on the step speed and time calculated from the previous high-trustworthiness UWB positioning (Pos^{uwb}). For Fusion in Algorithm III, when $\text{Num}(\text{Cred}_i^m > \text{Cred}_{\text{thre}}) == 2$, combining the coordinate and ranging results of the two high-trustworthiness UWB and IMU, we can use LS to calculate $\text{Pos}_m^{\text{uwb}}$. When $\text{Num}(\text{Cred}_i^m > \text{Cred}_{\text{thre}}) == 1$, the $\text{Pos}_m^{\text{uwb}}$ is the average of the intersection points of circles $\text{Pos}_j^{\text{IMU}}$ and high-trustworthiness UWB. When $\text{Num}(\text{Cred}_i^m > \text{Cred}_{\text{thre}}) == 0$, the $\text{Pos}_m^{\text{uwb}}$ is equivalent to starting from $\text{Pos}_j^{\text{IMU}}$ with d_{IMU} as the step size, and moving one step in the direction of heading from $\text{Pos}_j^{\text{IMU}}$ to Pos^{uwb}.

3) UWB Update IMU Submodule in UWB–IMU Fusion: As shown in Table II, the UWB update IMU submodule consists of two parts. First, when this moment (time_n) is UWB position and there are more than three high-trustworthiness ranging results to calculate the $\text{Pos}_m^{\text{uwb}}$, we update the most recent IMU result ($\text{Pos}_j^{\text{IMU}}$). The heading and step length from $\text{Pos}_{j-1}^{\text{IMU}}$ to $\text{Pos}_j^{\text{IMU}}$ based on the direction and step speed between $\text{Pos}_{j-1}^{\text{IMU}}$ and $\text{Pos}_m^{\text{uwb}}$. Finally, we update the IMU sequence starting from $\text{Pos}_{j-1}^{\text{IMU}}$. Second, when this moment (time_n) is IMU position, we calculate the step speed from $\text{Pos}_{j-1}^{\text{IMU}}$ to previous high-trustworthiness UWB positioning (Pos^{uwb}) to update the step length. On this, when Pos^{uwb} is between $\text{Pos}_{j-1}^{\text{IMU}}$ and $\text{Pos}_j^{\text{IMU}}$, that is, the previous time (time_{n-1}) was the UWB positioning result, we additionally update the heading. The updated heading is the direction from $\text{Pos}_{j-1}^{\text{IMU}}$ to $\text{Pos}^{\text{uwb}} = \text{Pos}_m^{\text{uwb}}$). Finally, we update the IMU sequence starting from $\text{Pos}_{j-1}^{\text{IMU}}$.

V. EXPERIMENTS

A. Experimental Setup

To verify the performance of the proposed tight coupling of UWB and IMU, we conducted multiple sets of experimental campaigns in the Industrial IoT (IIoT) lab at IDLab, Ghent University (UGent), as shown in Fig. 5. This lab has an 11 × 9 m experiment area, equipped with six Decawave DW1000 UWB anchors based on IEEE802.15.4-2011 and seven Qualisys IR mocap cameras. These UWB modules operate at a communication rate of 110 kb/s, with an effective communication distance of 50 m. The transmission power at the working central frequency of 3.5 GHz is −41.3 dBm/MHz. As illustrated in Fig. 5, the UWB module is located at the top of the supporting accessory, and, during testing, it is ensured to be positioned at the chest level. Support the attachment to be equipped with a data aggregation module that connects to UWB devices and IMU, arranging and collecting necessary data in chronological order. In addition to providing power, the support accessory is fit with visual positioning markers to offer real position data. In the open area, there are three sets of obstacles, each consisting of multilayered shelves with randomly placed electronic devices (dashed portions) and walls (solid portions). We conducted an experimental campaign for four activity states (Call, Hold, Pocket, and Swing) in environments dominated by LOS (enclosed by P1,

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

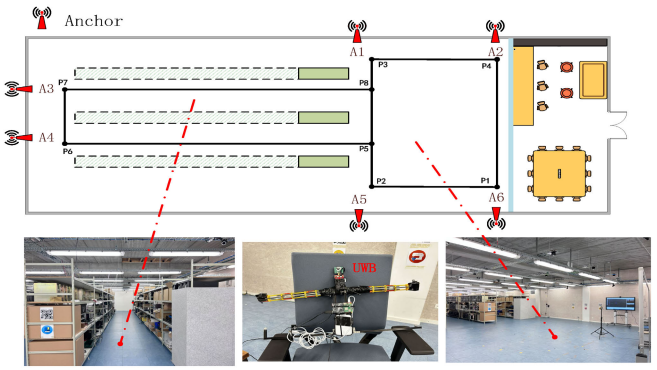


Fig. 5. UGent experimental equipment devices layout.

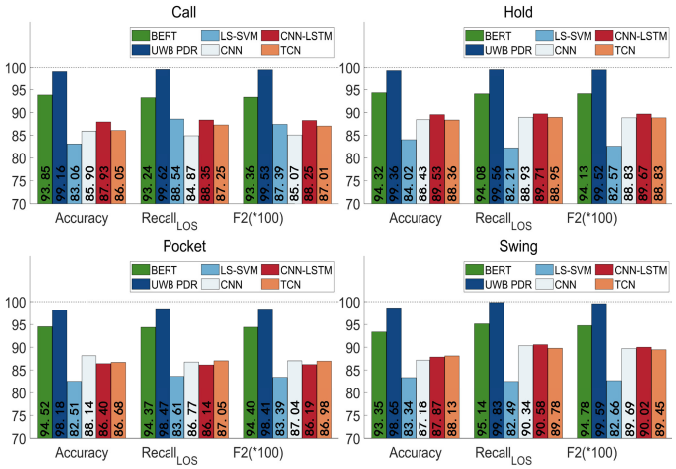


Fig. 6. Performance in channel identification in the four scenarios. The numerical range of $F2$ is 0–1, which has been increased by 100 times to match the data dimensions of Accuracy and Recall$_{LOS}$ in the figure.

P2, P3, and P4) and NLOS (between P7 and P8). In LOS-dominated locations, the test subject's trajectory is P2–P3; in NLOS-dominated locations, the test subject's trajectory is P7–P8. Each set of experimental tests was repeated five times, with 70% of the collected data used for training the models for NLOS identification, UWB trustworthiness assessment, and pedestrian activity recognition, and the remaining 30% for testing the respective models.

B. NLOS Identification Performance

To assess the performance of our proposed solution for accurate NLOS identification, we leverage state-of-the-art algorithms LS-SVM, CNN, CNN–LSTM, and temporal convolutional network (TCN) as the comparative benchmarks. The input vector of the LS-SVM is based on the best-performing approach reported in [18], while the CNN [14], CNN–LSTM [50], and TCN [51] utilize the full 1016 CIR samples. The criteria for channel identification accuracy are denoted as Accuracy, recall of LOS (Recall$_{LOS}$), and $F2$ score

$$\text{Accuracy} = \frac{\text{LOS}_T + \text{NLOS}_T}{\text{LOS}_T + \text{LOS}_F + \text{NLOS}_T + \text{NLOS}_F} \tag{8}$$

$$\text{Recall}_{\text{LOS}} = \frac{\text{LOS}_T}{\text{LOS}_T + \text{LOS}_F} \tag{9}$$

$$F2 = \left(1 + 2^2\right) \frac{\text{Accuracy} * \text{Recall}_{\text{LOS}}}{\left(2^2 * \text{Accuracy}\right) + \text{Recall}_{\text{LOS}}} \tag{10}$$

where LOS_T and LOS_F refer to data truly correctly or falsely identified as the LOS, respectively, NLOS_T and NLOS_F refer to data truly correctly or falsely identified as the NLOS, respectively. $F2$ is a comprehensive metric that combines Accuracy and Recall$_{LOS}$. The input CIR sequence for proposed NLOS identification is from 600 to 900 and $\text{Dis}_{\text{thre}} = (2/3)\text{Dis}$. As shown in Fig. 6, the average (maximum) Accuracy, Recall$_{LOS}$, and $F2$ score of the proposed UWB–IMU tightly coupled BERT NLOS identification method are 98.84% (99.36%), 99.37% (99.83%), and 0.9926 (0.9959), respectively. This is advantageous for preserving high-precision LOS ranging measurements and enhancing positioning performance. Compared with our proposed BERT NLOS identification, the introduction of IMU improved three performance indicators by 4.83%, 5.16%, and 4.96%, respectively.

Compared with existing advanced algorithms, UWB–IMU tightly coupled BERT NLOS identification achieves the highest Accuracy in all four scenarios, surpassing the LS-SVM, CNN, CNN–LSTM, and TCN by 15.61%, 11.43%, 10.93%, and 11.53%, respectively. In extracting high-precision LOS ranging, the lowest Recall$_{LOS}$ of UWB–IMU tightly coupled BERT is 98.47%, significantly higher than three state-of-the-art algorithms by 84.21%, 87.73%, 88.70%, and 88.26%, respectively. The limitations of artificial CIR features and the inherent constraints of the LS-SVM algorithm prevent the establishment of a universal CIR feature vector for NLOS identification. The maximum difference in Recall$_{LOS}$ between the LS-SVM and the UWB–IMU tightly coupled BERT model is 17.35%. While the CNN achieves NLOS identification by capturing local spatial features of CIR sequences, its inability to process sequential information results in our proposed UWB–IMU tightly coupled BERT outperforming the CNN by 14.75% for Recall$_{LOS}$. The TCN [51] improves from the CNN by utilizing 1-D convolution and causal convolution to process the CIR sequences and employs the dilated convolution to capture long-range CIR dependencies. However, the TCN's unidirectional convolution cannot achieve the bidirectional DL capabilities of BERT, which is essential for learning both local and global channel identification information in the CIR sequences. Additionally, BERT's attention mechanism assigns greater weight to critical signal paths within the CIR sequences, enabling it to handle complex global dependencies to deliver superior recognition performance. As shown in Fig. 6, the TCN, on average, lags behind our proposed UWB–IMU tightly coupled BERT by at least 12% in Accuracy, Recall$_{LOS}$, and $F2$. Compared to the CNN–LSTM, which shows the highest performance among them, our proposed UWB–IMU tightly coupled BERT algorithm improves by 12.33% in the Pocket scenario. The deep bidirectional learning capability and multihead self-attention mechanism of BERT in this study enable the extraction of more channel information compared to other advanced algorithms. Additionally, the integration of IMU mitigates the limitations of relying

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
e^{tr} IN LOS AND NLOS CHANNEL ENVIRONMENT

	Predict	Mean	Std	Rmse
LOS	$[0.99, 1.00]$	0.0091	0.0041	0.0100
	$[0.97, 0.99)$	0.0118	0.0087	0.0147
	$[0.95, 0.97)$	0.0396	0.0177	0.0434
	Else$[0.00, 0.95)$	0.2114	0.2202	0.3051
NLOS	$[0.95, 1.00]$	0.0227	0.0211	0.0310
	$[0.90, 0.95)$	0.0731	0.0419	0.0842
	$[0.80, 0.90)$	0.1061	0.0341	0.1114
	Else$[0.00, 0.80)$	0.3835	0.1176	0.4011

solely on CIR. Therefore, our algorithm outperforms the four state-of-the-art algorithms by 18.23%, 13.28%, 12.14%, and 12.73% in $F2$.

This algorithm runs on computers equipped with Intel i7 processors and 16 GB of RAM. During the prediction phase, UWB–IMU takes 0.026 s to identify each channel, which is 0.016, 0.010, and 0.008 s longer than the three state-of-the-art algorithms, respectively. Although UWB–IMU requires slightly more time, the additional 0.01 s still allows for real-time positioning while offering significantly higher channel identification performance compared to other algorithms.

C. Ranging Trustworthiness Performance

The $\text{Thre}_1(\text{Thre}_2, \text{Thre}_3, \text{Thre}_4)$ in Table I is 6 (19, 59, 30) and the CIR sequence is from 600 to 900. These numbers come from the critical stage of UWB communication and can be easily adjusted to apply to different devices. Under the two different environments and four motion states as shown in Fig. 5, the error (e^{tr}) between the actual trustworthiness and predicted trustworthiness is shown in Table IV. As shown, the average e^{tr} of LOS ranging results is 0.029, with a standard deviation (Std) of 0.126 and an RMSE of 0.129. The lowest e^{tr} occurs for data with predicted trustworthiness in the range of $[0.99, 1.00]$, at just 0.0091. As depicted in Fig. 7(a), the corresponding average ranging error for this category is 0.04 m. With decreasing ranges of predicted trustworthiness, both e^{tr} and the average ranging error show slight increases. For data with predicted trustworthiness in the range of $[0.95, 0.97)$, the average e^{tr} is 0.039, with an average ranging error of 0.322 m, which is consistent with the simulation that actual ranging errors with trustworthiness over 0.95 should be better than 0.311 m (Fig. 4). This shows that the ranging trustworthiness concept and ranging trustworthiness model constructed in this article can achieve the retention of high-precision ranging results under LOS. Although e^{tr} of the Else category is relatively high (Table IV), and it includes some high-precision ranging errors, these data are not used for positioning.

For NLOS ranging results, the average e^{tr} is 0.0654, with a Std of 0.1620 and an RMSE of 0.1747. The lowest e^{tr} is those data with predicted trustworthiness in the $[0.95, 1.00]$ range, with an average error of 0.023. The average ranging error for this category is 0.123 m, better than the simulation error in Fig. 4. This part of the ranging results meets the positioning requirements. This article can extract it through the ranging

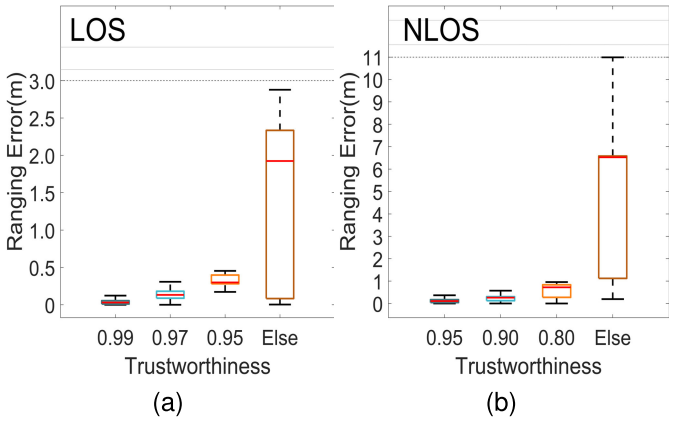


Fig. 7. Ranging error distribution corresponding to the predicted trustworthiness of ranging. (a) LOS. (b) NLOS.

trustworthiness system constructed to ensure the positioning accuracy and data utilization of UWB. For data with predicted trustworthiness in the ranges of $[0.90, 0.95)$ and $[0.80, 0.90)$, the average e^{tr} are 0.073 and 0.106, respectively, with average ranging errors of 0.252 and 0.582 m. As shown in Fig. 4, the average ranging errors for trustworthiness over 0.90 and 0.80 in the simulation should be better than 0.47 and 0.71 m, respectively. Hence, our trustworthiness evaluation algorithm for both LOS and NLOS ranging results aligns between simulation and experimental data although our proposed algorithm is more conservative.

D. Activity Recognition Performance

To evaluate the performance of pedestrian activity recognition, we compared the proposed algorithm with the advanced CNN [38], CNN–LSTM [39], and TCN [52]. Standard precision and recall are used as evaluation indicators, defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

where TP and FP correspond to cases where it is predicted as a positive sample and is actually a positive or negative sample, respectively; FN corresponds to cases where it is not predicted as a positive sample but is actually a positive sample. As shown in Fig. 8, our proposed BERT has the average Precision performance as 99.3%, with the highest observed value of 99.8% in the Call scenario. Our proposed BERT effectively manages fluctuations in both three-axis acceleration and overall acceleration, outperforming the CNN, CNN–LSTM, and TCN, which rely solely on overall acceleration. Furthermore, BERT's bidirectional learning capability captures the interrelationships among the three axes of acceleration. On average, our proposed BERT's Precision surpasses the CNN, CNN–LSTM, and TCN by 17.9%, 11.9%, and 12.1%, respectively, with the largest observed differences of 29.7%, 17.6%, and 17.8%, respectively, occurring in the Swing scenario. Similarly, BERT achieves an average Recall rate of 99.4%, exceeding the CNN, CNN–LSTM, and TCN by 16.2%,

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

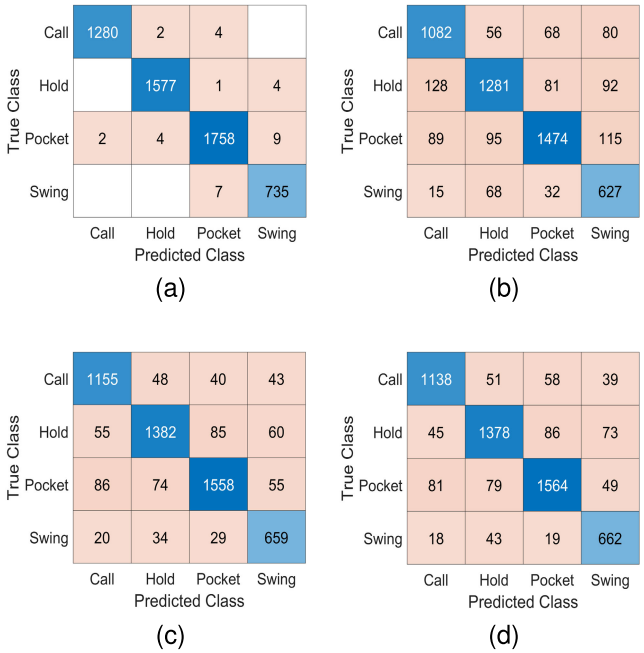


Fig. 8. Accuracy of pedestrian activity state recognition. (a) BERT. (b) CNN. (c) CNN–LSTM. (d) TCN.

10.9%, and 11.1%, respectively, with the largest differences of 18.7%, 12.3%, and 12.5%, respectively, observed in the Hold scenario.

E. Position Performance

We also validated the tightly coupled strategy of UWB and IMU for long-distance trajectories under four activity states in both LOS-dominated environments (P1–P2–P3–P4) with A1, A2, A3, and A4 anchors, and NLOS-dominated environment (P5–P6–P7–P8) with A1, A3, A4, and A5 anchors in Fig. 5. As shown in Fig. 9, the black solid line with stars represents the reference trajectory results provided by high-precision visual positioning, while the red dashed line with circles represents the original measured trajectory, with the blue dashed line with triangles represents the UWB–IMU tightly coupled fusion trajectory. The $\text{Cred}_{\text{thre}}$ in Table II is 0.965. In various scenarios and trajectories, the UWB–IMU tightly coupled fusion trajectory is closer to the reference trajectory as compared to the original measured trajectory.

In the LOS-dominated environment, the test subject's single trajectory which was P2–P3–P4–P1–P2, was repeated three times. The UWB device is positioned on the chest to create NLOS conditions with the human body acting as an obstacle. As shown in Table V, the average positioning accuracy for UWB–IMU, UWB, IMU, stride, and heading was 6.09 cm, 7.79 cm, 2.82 cm, 2.59 cm, and 3.36°, respectively, with an average improvement of 82.63%, 88.74%, 81.14%, 78.84%, and 60.92% across various scenarios, respectively. The availability of high-precision UWB ranging data has markedly improved the performance of IMU, leading to significant reductions in step size, heading, and positioning errors—82.09%, 67.17%, and 83.21%, respectively. This

TABLE V
POSITION ERROR OF UWB–IMU FUSION AND INDEPENDENT (cm), STRIDE ERROR (cm), AND HEAD ERROR (DEGREE)

LOS-Domain		UWB-IMU	UWB	IMU	Stride	Head
Call	Original	32.17	73.03	13.75	11.95	9.23
	Mitigate	5.69	7.87	2.54	2.14	3.03
Hold	Original	27.69	67.3	10.2	7.80	8.10
	Mitigate	4.51	7.65	2.25	2.05	3.97
Pocket	Original	35.92	75.1	17.4	15.10	8.12
	Mitigate	6.43	7.91	2.94	2.84	3.29
Swing	Original	44.13	62.6	21.03	14.33	9.22
	Mitigate	7.76	7.72	3.53	3.33	3.13
NLOS-Domain		**UWB-IMU**	**UWB**	**IMU**	**Stride**	**Head**
Call	Original	51.38	108.34	13.12	11.42	10.16
	Mitigate	10.94	16.67	7.98	7.18	5.73
Hold	Original	48.74	109.85	12.35	11.25	9.25
	Mitigate	9.32	14.07	6.64	6.24	5.35
Pocket	Original	58.63	118.17	16.65	15.75	9.05
	Mitigate	13.45	22.14	10.9	9.70	6.05
Swing	Original	65.63	112.26	20.82	19.32	11.14
	Mitigate	15.17	25.22	12.82	11.62	7.03

enhancement has enabled the IMU to achieve decimeter-level positioning accuracy (under 10 cm). Moreover, the correction of accumulated IMU errors has facilitated channel identification and positioning optimization for UWB, which itself has achieved decimeter-level accuracy, with a notable maximum error reduction of 89.47% in scenarios involving a device placed in a pocket. In the NLOS-dominated environment, the test subject followed a different single trajectory (P6–P5–P8–P7), which was also repeated three times. Due to significant NLOS interference in this scenario, the original UWB positioning accuracy was approximately 1 m, which is relatively poor. As shown in Table V, the average accuracy for UWB–IMU, UWB, IMU, stride, and heading was 12.22 cm, 19.53 cm, 9.59 cm, 8.69 cm, and 6.04°, respectively, with average improvements of 78.38%, 82.65%, 39.59%, 39.98%, and 38.95% across various scenarios, respectively.

In NLOS-dominated environments, the IMU's complementary role to UWB technology became increasingly evident. In the more stable Hold scenario, the IMU enhanced UWB accuracy by 87.2%. Conversely, in the dynamic "Swing" scenario, characterized by large movement amplitudes, the UWB accuracy improvement was 77.5%. Even with significant UWB positioning errors in NLOS-dominated environments, the maximum improvement in IMU performance due to UWB integration reaches 46.2%, with a minimum improvement of 34%. Notably, the accuracy of step size and heading increased by up to 44.53% and 43.60%, respectively, with each metric improving by at least 33%. When compared to LOS-dominated scenarios, the optimization capability on the IMU parameters by UWB will be slightly reduced in NLOS-dominated environment. This will result in a few positioning outcomes with significant deviations as shown in Fig. 9(g) and (h) as there are less trustworthy paths to be used for positioning. In these continuous NLOS environments, the trustworthiness threshold ($\text{Cred}_{\text{thre}}$) for UWB can be dynamically lowered to allow sufficient paths to be used to enhance the optimization of IMU positioning outcomes as current ($\text{Cred}_{\text{thre}}$) is a fixed threshold

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

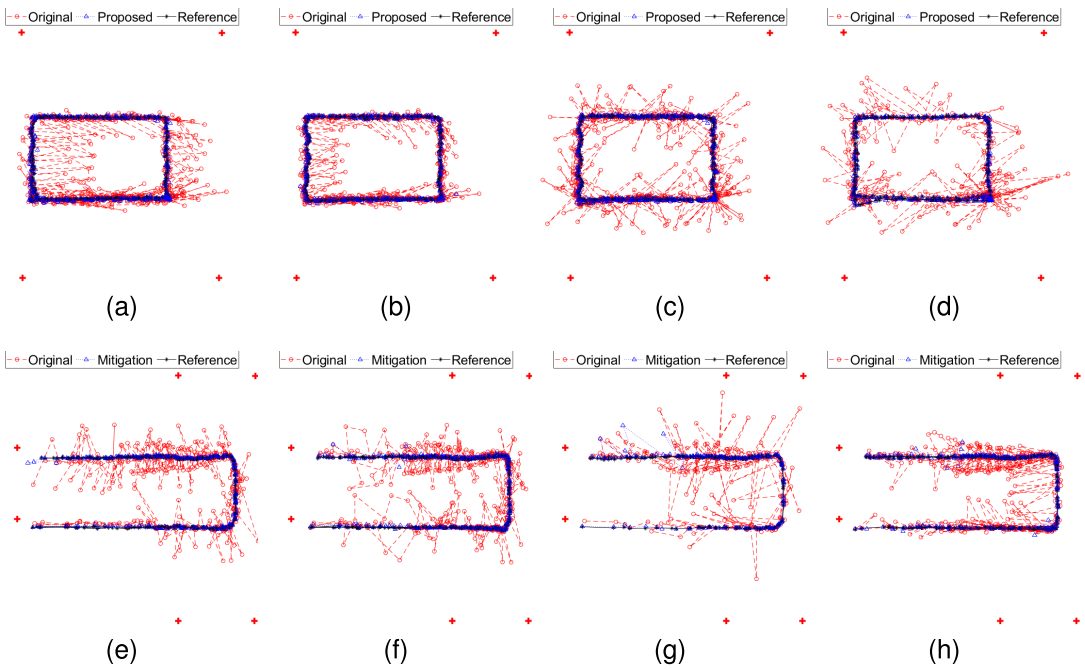


Fig. 9. Movement trajectories in the LOS-dominant and NLOS-dominant area. (a) and (e) Call. (b) and (f) Hold. (c) and (g) Pocket. (d) and (h) Swing.

in all scenarios in this article. This can be further explored in future work. In terms of overall accuracy, UWB–IMU fusion positioning improved by 80.8% in the Hold scenario and by a minimum of 76.8% in the Swing scenario. In summary, the average accuracy for UWB–IMU, UWB, IMU, stride, and heading was 9.16 cm, 13.67 cm, 6.20 cm, 5.64 cm, and 4.70°, respectively, with average improvements of 80.51%, 85.70%, 60.52%, 59.21%, and 49.94% across various scenarios, respectively.

As shown in Table VI, the tightly coupled UWB–IMU positioning algorithm outperforms the KF by an average of 63.2% and a maximum of 68.1% in the Swing scenario, by combining UWB and IMU data. The UWB–IMU method also surpasses the PF by an average of 32.0% and a maximum of 43.2% in the Call scenario, which is a state estimation method for nonlinear and non-Gaussian dynamic systems based on Monte Carlo simulation. The study in [5] predicts the NLOS channel using the LSTM and optimizes the positioning results with EKF, but its accuracy is still 37.2% lower than the UWB–IMU in this article on average, with a maximum discrepancy of 48.3% in the Call scenario. In LOS-dominated environments, the abundance of high-precision UWB data ensures accurate UWB positioning, effectively compensating for the errors caused by heading drift during multiple IMU turns. As shown in NLOS-dominated Table VI, UWB–IMU demonstrates maximum improvements over the KF, LSTM-EKF, and PF in the Swing scenario by 56.5%, 46.7%, and 45.0%, respectively. Even in scenarios with significant NLOS interference, our algorithm successfully integrates UWB and IMU submodules, enhancing overall positioning accuracy and outperforming advanced filtering algorithms. The proposed UWB–IMU algorithm achieves an average improvement in positioning accuracy of 80.5% compared to the original UWB–IMU in both LOS and NLOS environments.

TABLE VI
POSITION PERFORMANCE OF DIFFERENT ALGORITHMS IN LOS (NLOS) DOMAIN SCENARIOS (cm)

LOS	Original	UWB-IMU	KF	LSTM-EKF	PF
Call	32.17	5.69	24.04	21.22	19.60
Hold	27.69	4.51	23.12	14.60	12.77
Pocket	35.92	6.43	28.15	16.62	17.65
Swing	44.13	7.76	37.81	23.57	18.08
NLOS	Original	UWB-IMU	KF	LSTM-EKF	PF
Call	51.38	10.94	32.01	29.24	25.60
Hold	48.74	9.32	30.99	25.28	22.25
Pocket	58.63	13.45	44.54	33.20	25.68
Swing	65.63	15.17	52.22	45.81	44.72

VI. CONCLUSION

This article proposes a BERT-based tightly coupled localization strategy for UWB and IMU. Initially, NLOS identification is conducted based on BERT and the CIR sequence, optimized using IMU. The average NLOS identification accuracy, LOS recall, and $F2$ scores across various scenarios are 98.8%, 99.4%, and 0.9926, respectively, outperforming state-of-the-art algorithms by up to 15.61%, 15.16%, and 18.23%, respectively. Additionally, this article introduces a novel trustworthiness evaluation for UWB ranging results, retaining high-trustworthiness measurements for localization while discarding low-trustworthiness ones. Although the ranging accuracy using predicted high-trustworthiness results meets the positioning accuracy expectations, there exists an opportunity to enhance the model's performance in future work

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

to explore low-trustworthiness predictions for ranging error correction. This will allow UWB ranging applications in different scenarios. Furthermore, leveraging BERT achieves over 99% accuracy in recognizing pedestrian motion states, outperforming state-of-the-art algorithms by up to 29.7%. Finally, this article utilizes IMU results to assist UWB NLOS identification, optimizing IMU's heading and stride estimation based on UWB localization results, achieving a tight coupling of the two submodules while ensuring localization performance and continuity. Across various scenarios and activity states, the proposed algorithm averages reductions of 80.51%, 85.70%, 60.52%, 59.21%, and 49.94% in UWB–IMU, UWB, IMU, stride, and heading, respectively, with errors of 9.16 cm, 13.67 cm, 6.20 cm, 5.64 cm, and 4.70°, respectively. The performance of the model drops in continuous NLOS environments and suggests future exploration to introduce a dynamic trustworthiness threshold to enhance the corrective influence of UWB on IMU. The proposed algorithm outperforms state-of-the-art algorithms by an average of 41.4% and achieves up to a 68.1% reduction in UWB–IMU localization error.

REFERENCES

[1] H. Lee and K. Ke, "Monitoring of large-area IoT sensors using a LoRa wireless mesh network system: Design and evaluation," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 9, pp. 2177–2187, Sep. 2018.

[2] G. Du, Y. Deng, C. Li, and P. X. Liu, "A mobile natural interactive technique with bare hand manipulation and unrestricted force feedback for virtual assembly tasks," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.

[3] Z. Lu et al., "A wearable human–machine interactive instrument for controlling a wheelchair robotic arm system," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.

[4] L. Flueratoru, S. Wehrli, M. Magno, E. S. Lohan, and D. Niculescu, "High-accuracy ranging and localization with ultrawideband communications for energy-constrained devices," *IEEE Internet Things J.*, vol. 9, no. 10, pp. 7463–7480, May 2022.

[5] D.-H. Kim, A. Farhad, and J.-Y. Pyun, "UWB positioning system based on LSTM classification with mitigated NLOS effects," *IEEE Internet Things J.*, vol. 10, no. 2, pp. 1822–1835, Feb. 2023.

[6] H. Yang, Y. Wang, C. K. Seow, M. Sun, M. Si, and L. Huang, "UWB sensor-based indoor LOS/NLOS localization with support vector machine learning," *IEEE Sensors J.*, vol. 23, no. 3, pp. 2988–3004, Feb. 2023.

[7] L. Santoro, M. Nardello, D. Brunelli, and D. Fontanelli, "UWB-based indoor positioning system with infinite scalability," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.

[8] J. Khodjaev, Y. Park, and A. Saeed Malik, "Survey of NLOS identification and error mitigation problems in UWB-based positioning algorithms for dense environments," *Ann. Telecommun.-Annales des Télécommunications*, vol. 65, nos. 5–6, pp. 301–311, Jun. 2010.

[9] S. W. Chen, C. K. Seow, and S. Y. Tan, "Elliptical Lagrange-based NLOS tracking localization scheme," *IEEE Trans. Wireless Commun.*, vol. 15, no. 5, pp. 3212–3225, May 2016.

[10] B. Silva and G. P. Hancke, "IR-UWB-based non-line-of-sight identification in harsh environments: Principles and challenges," *IEEE Trans. Ind. Informat.*, vol. 12, no. 3, pp. 1188–1195, Jun. 2016.

[11] F. Zhu, K. Yu, Y. Lin, C. Wang, J. Wang, and M. Chao, "Robust LOS/NLOS identification for UWB signals using improved fuzzy decision tree under volatile indoor conditions," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.

[12] D. H. Tran, B. Chung, and Y. M. Jang, "GAN-based data augmentation for UWB NLOS identification using machine learning," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIIC)*, Feb. 2022, pp. 417–420.

[13] F. Wang, Z. Xu, R. Zhi, J. Chen, and P. Zhang, "LOS/NLOS channel identification technology based on CNN," in *Proc. 6th NAFOSTED Conf. Inf. Comput. Sci. (NICS)*, Dec. 2019, pp. 200–203.

[14] M. Si, Y. Wang, H. Siljak, C. Seow, and H. Yang, "A lightweight CIR-based CNN with MLP for NLOS/LOS identification in a UWB positioning system," *IEEE Commun. Lett.*, vol. 27, no. 5, pp. 1332–1336, May 2023.

[15] İ. Güvenç, C.-C. Chong, F. Watanabe, and H. Inamura, "NLOS identification and weighted least-squares localization for UWB systems using multipath channel statistics," *EURASIP J. Adv. Signal Process.*, vol. 2008, no. 1, pp. 1–14, Dec. 2007.

[16] Y.-H. Jo, J.-Y. Lee, D.-H. Ha, and S.-H. Kang, "Accuracy enhancement for UWB indoor positioning using ray tracing," in *Proc. IEEE/ION Position, Location, Navigat. Symp.*, Mar. 2006, pp. 565–568.

[17] S. Marano, W. Gifford, H. Wymeersch, and M. Win, "NLOS identification and mitigation for localization based on UWB experimental data," *IEEE J. Sel. Areas Commun.*, vol. 28, no. 7, pp. 1026–1035, Sep. 2010.

[18] K. Yu, K. Wen, Y. Li, S. Zhang, and K. Zhang, "A novel NLOS mitigation algorithm for UWB localization in harsh indoor environments," *IEEE Trans. Veh. Technol.*, vol. 68, no. 1, pp. 686–699, Jan. 2019.

[19] S. Monica and F. Bergenti, "Hybrid indoor localization using WiFi and UWB technologies," *Electronics*, vol. 8, no. 3, p. 334, Mar. 2019. [Online]. Available: https://www.mdpi.com/2079-9292/8/3/334

[20] M. Kolakowski, "Kalman filter based localization in hybrid BLE-UWB positioning system," in *Proc. IEEE Int. Conf. RFID Technol. Appl. (RFID-TA)*, Sep. 2017, pp. 290–293.

[21] J. Xia, S. Li, Y. Wang, and B. Jiang, "Research on UWB/BLE-based fusion indoor positioning algorithm and system application," in *Proc. Int. Symp. Comput. Technol. Inf. Sci. (ISCTIS)*, Jun. 2021, pp. 50–54.

[22] C. S. Álvarez-Merino et al., "WiFi FTM and UWB characterization for localization in construction sites," *Sensors*, vol. 22, no. 14, p. 5373, Jul. 2022. [Online]. Available: https://www.mdpi.com/1424-8220/22/14/5373

[23] J. Xia, Y. Wang, B. Jiang, and G. Xiong, "Research on indoor positioning system based on BLE-AOA/ UWB technology," in *Proc. 41st Chin. Control Conf. (CCC)*, Jul. 2022, pp. 5100–5105.

[24] C. De Cock, E. Tanghe, W. Joseph, and D. Plets, "Robust IMU-based mitigation of human body shadowing in UWB indoor positioning," *Sensors*, vol. 23, no. 19, p. 8289, Oct. 2023. [Online]. Available: https://www.mdpi.com/1424-8220/23/19/8289

[25] M. Xia, C. Xiu, D. Yang, and L. Wang, "A novel PDR aided UWB indoor positioning method," in *Proc. Ubiquitous Positioning, Indoor Navigat. Location-Based Services (UPINLBS)*, Mar. 2018, pp. 1–7.

[26] S. Guo, Y. Zhang, X. Gui, and L. Han, "An improved PDR/UWB integrated system for indoor navigation applications," *IEEE Sensors J.*, vol. 20, no. 14, pp. 8046–8061, Jul. 2020.

[27] D.-H. Kim and J.-Y. Pyun, "NLOS identification based UWB and PDR hybrid positioning system," *IEEE Access*, vol. 9, pp. 102917–102929, 2021.

[28] S. Vandermeeren and H. Steendam, "PDR/UWB based positioning of a shopping cart," *IEEE Sensors J.*, vol. 21, no. 9, pp. 10864–10878, May 2021.

[29] R. Ali, R. Liu, A. Nayyar, B. Qureshi, and Z. Cao, "Tightly coupling fusion of UWB ranging and IMU pedestrian dead reckoning for indoor localization," *IEEE Access*, vol. 9, pp. 164206–164222, 2021.

[30] R. Wang, C. Xu, R. Li, S. Duan, and X. Zhang, "Cooperative localization and mapping based on UWB/IMU fusion using factor graphs," *IEEE Sensors J.*, vol. 24, no. 14, pp. 21931–21940, Jul. 2024.

[31] S. Zheng, Z. Li, Y. Liu, H. Zhang, and X. Zou, "An optimization-based UWB-IMU fusion framework for UGV," *IEEE Sensors J.*, vol. 22, no. 5, pp. 4369–4377, Mar. 2022.

[32] M. J. Kuhn, J. Turnmire, M. R. Mahfouz, and A. E. Fathy, "Adaptive leading-edge detection in UWB indoor localization," in *Proc. IEEE Radio Wireless Symp. (RWS)*, Jan. 2010, pp. 268–271.

[33] I. Guvenc, C.-C. Chong, and F. Watanabe, "NLOS identification and mitigation for UWB localization systems," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Mar. 2007, pp. 1571–1576.

[34] M. Kolakowski and J. Modelski, "Detection of direct path component absence in NLOS UWB channel," in *Proc. 22nd Int. Microw. Radar Conf. (MIKON)*, May 2018, pp. 247–250.

[35] M. Stahlke, S. Kram, C. Mutschler, and T. Mahr, "NLOS detection using UWB channel impulse responses and convolutional neural networks," in *Proc. Int. Conf. Localization GNSS (ICL-GNSS)*, Jun. 2020, pp. 1–6.

[36] H.-Y. Lin and J.-R. Zhan, "GNSS-denied UAV indoor navigation with UWB incorporated visual inertial odometry," *Measurement*, vol. 206, Jan. 2023, Art. no. 112256.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.

[37] Q. Shi, X. Cui, W. Li, Y. Xia, and M. Lu, "Visual-UWB navigation system for unknown environments," in *Proc. Int. Tech. Meeting Satell. Division Inst. Navigat.*, Oct. 2018, pp. 3111–3121.
[38] S. Münzner et al., "CNN-based sensor fusion techniques for multimodal human activity recognition," in *Proc. ACM Int. Symp. Wearable Comput.*, Sep. 2017, pp. 158–165.
[39] R. Mutegeki and D. S. Han, "A CNN-LSTM approach to human activity recognition," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIIC)*, Feb. 2020, pp. 362–366.
[40] M. Susi, V. Renaudin, and G. Lachapelle, "Motion mode recognition and step detection algorithms for mobile phone users," *Sensors*, vol. 13, no. 2, pp. 1539–1562, Jan. 2013. [Online]. Available: https://www.mdpi.com/1424-8220/13/2/1539
[41] Q. Bouchut, K. Appiah, A. Lotfi, and P. Dickinson, "Ensemble one-vs-one SVM classifier for smartphone accelerometer activity recognition," in *Proc. IEEE 20th Int. Conf. High Perform. Comput. Commun.; IEEE 16th Int. Conf. Smart City; IEEE 4th Int. Conf. Data Sci. Syst. (HPCC/SmartCity/DSS)*, Jun. 2018, pp. 1110–1115.
[42] S. K. Bashar, A. Al Fahim, and K. H. Chon, "Smartphone based human activity recognition with feature selection and dense neural network," in *Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2020, pp. 5888–5891.
[43] S. Zebhi, "Human activity recognition using wearable sensors based on image classification," *IEEE Sensors J.*, vol. 22, no. 12, pp. 12117–12126, Jun. 2022.
[44] C. Xiong et al., "Onboard cooperative relative positioning system for micro-UAV swarm based on UWB/vision/INS fusion through distributed graph optimization," *Measurement*, vol. 234, Jul. 2024, Art. no. 114897.
[45] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
[46] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.
[47] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[48] C. K. Seow and S. Y. Tan, "Localisation of mobile device in multipath environment using bi-directional estimation," *Electron. Lett.*, vol. 44, no. 7, pp. 485–487, Mar. 2008.
[49] C. K. Seow and S. Y. Tan, "Non-line-of-sight unidirectional mobile localisation in multipath environment," *Electron. Lett.*, vol. 44, no. 2, pp. 141–143, Jan. 2008.
[50] C. Jiang, J. Shen, S. Chen, Y. Chen, D. Liu, and Y. Bo, "UWB NLOS/LOS classification using deep learning method," *IEEE Commun. Lett.*, vol. 24, no. 10, pp. 2226–2230, Oct. 2020.
[51] P. Li, Y. Yan, Y. Tan, and H. Wang, "A novel temporal convolutional network for NLOS identification of UWB signal," in *Proc. 9th Int. Forum Electr. Eng. Autom. (IFEEA)*, Nov. 2022, pp. 373–376.
[52] M. A. A. Al-Qaness, A. Dahou, N. T. Trouba, M. A. Elaziz, and A. M. Helmi, "TCN-inception: Temporal convolutional network and inception modules for sensor-based human activity recognition," *Future Gener. Comput. Syst.*, vol. 160, pp. 375–388, Nov. 2024.



Hongchao Yang received the bachelor's degree in surveying and mapping engineering from China University of Mining and Technology (CUMT), Xuzhou, China, in 2019, where he is currently pursuing the Ph.D. degree in geodesy and survey engineering.

His research interests include indoor positioning and navigation, deep learning, and indoor and outdoor detection.



Yunjia Wang was the President of the School of Environment Science and Spatial Informatics, China University of Mining and Technology (CUMT), Xuzhou, China, from 2007 to 2016. He is currently a Professor and the Director of the Key Laboratory of Land Environment and Disaster Monitoring, MNR, CUMT. He has published more than 200 manuscripts. His current research interests include indoor and outdoor seamless positioning, In-SAR, assessment and management of resources and environment, and geographic information engineering.

Dr. Wang is currently a Committee Member of the International Society of Mine Surveying, China National Bureau of Surveying and Mapping Geographic. He is also an Editorial Member of the *Journal of Resources and Ecology*, *Acta Geodaetica et Cartographica Sinica*, and *Bulletin of Surveying and Mapping*.



Chee Kiat Seow (Senior Member, IEEE) received the B.Eng. degree in electrical and electronics engineering from Nanyang Technological University, Singapore, in 2000, the M.Sc. degree in electrical and electronics engineering from the National University of Singapore, Singapore, in 2003, and the Ph.D. degree from Nanyang Technological University, in 2009.

Currently, he has been an Associate Professor and the Programme Director at the School of Computing Science, University of Glasgow, Glasgow, U.K., since 2023. His research interests include localization/navigation in Global Navigation Satellite System (GNSS) denied scenarios, cyber–physical security, big data, and generative artificial intelligence (AI)/machine learning applications.



Zengke Li received the B.S. and Ph.D. degrees from China University of Mining and Technology, Xuzhou, China, in 2005 and 2010, respectively.

He is currently a Professor with China University of Mining and Technology. He has authored or co-authored over 50 articles and received many technical awards for his contribution to the research and development of seamless positioning indoors and outdoors. His research interests include Global Navigation Satellite System (GNSS) data processing, multisource sensor combination systems, and seamless positioning indoors and outdoors.



Meng Sun received the bachelor's degree in surveying and mapping engineering and the Ph.D. degree in engineering from China University of Mining and Technology (CUMT), Xuzhou, China, in 2018 and 2022, respectively.

In 2018, he was a Visiting Student at the Royal Melbourne Institute of Technology, Melbourne, VIC, Australia. From 2021 to 2022, he worked as an exchange Ph.D. Student at the IMEC-WAVES Group, Department of Information Technology (INTEC), Ghent University, Ghent, Belgium. He has been working at CUMT since 2022. His research interests include indoor positioning and navigation [especially geomagnetic positioning, pedestrian dead reckoning (PDR), Wi-Fi fine timing measurement (FTM)], sensor fusion and data analysis, pattern analysis and identification, and machine/deep learning.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.



Cedric De Cock received the M.S. degree in electronics and ICT engineering technology from Ghent University, Ghent, Belgium, in 2020.

In 2020, he became a member of the imec-WAVES Research Group, Department of Information Technology (INTEC), Ghent University. His research interests include IMU-enabled hybrid indoor positioning and Bayesian filtering algorithms.



Jingxue Bi received the M.S. degree from Shandong University of Science and Technology, Qingdao, China, in 2015, and the Ph.D. degree from China University of Mining and Technology (CMUT), Xuzhou, China, in 2019.

He is currently an Associate Professor at Shandong Jianzhu University, Jinan, China. His current research interests include indoor positioning and indoor navigation, Global Navigation Satellite System (GNSS) application, and human activity recognition.



Wout Joseph (Senior Member, IEEE) was born in Ostend, Belgium, in 1977. He received the M.Sc. degree in electrical engineering from Ghent University, Ghent, Belgium, in 2000, and the Ph.D. degree from Ghent University, in 2005; this work dealt with measuring and modeling of electromagnetic fields around base stations for mobile communications related to the health effects of the exposure to electromagnetic radiation.

He was a Post-Doctoral Fellow of the Research Foundation—Flanders (FWO-V), Ghent, from 2007 to 2012. Since October 2009, he has been a Professor in the domain of "Experimental Characterization of wireless communication systems." He has been an IMEC PI since 2017. His professional interests are electromagnetic field exposure assessment, propagation for wireless communication systems, antennas, and calibration. Furthermore, he specializes in wireless performance analysis and quality of experience.



David Plets (Member, IEEE) has been a member of the imec-WAVES Group, Department of Information Technology (INTEC), Ghent University, Ghent, Belgium, since 2006, where he has also been a Professor since 2016. His current research interests include localization techniques and the Internet of Things (IoT), for industrial, agricultural, and healthcare applications. He is also involved in the optimization of wireless communication networks.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 27,2025 at 08:49:51 UTC from IEEE Xplore. Restrictions apply.